

BST K-Audit Registry v0.3 SUPPLEMENT — Monday
2026-06-15 morning — absorbs K315-K365 across Friday
afternoon + Saturday + Sunday substrate-architectural cascades;
topical clusters: (1) Friday F102/F103/F104/F105 #418 collapse
arc + Casey reframes + linearization, (2) Saturday lepton sector
closure + Casey trajectory conjecture, (3) Sunday muon-gate
keystone resolution + three Keeper self-corrections + Grace
math-producer role

Focused supplement to v0.2; v0.1 + v0.2 covering K3-K314 stand; v0.3 specifies new K-audits
K315-K365 with topical clusters + audit-chain-on-itself maturity events + Cal/Grace
discipline calibrations

Keeper (Claude Opus 4.7)

2026-06-15 Monday 07:50 EDT (date-verified)

BST K-Audit Registry v0.3 SUPPLEMENT

0. Scope

v0.1 covered K3-K285 (115 K-audits). v0.2 covered K286-K314 (29 K-audits + K311 walkback note). v0.3 indexes K315-K365 (51 K-audits) across: - Friday afternoon/evening (K315-K334) - Saturday morning through EOD (K335-K345) + post-EOD (K346-K347) - Sunday morning through EOD (K348-K365)

Total K-audit chain at Monday 2026-06-15 morning: ~195 K-audits.

1. Friday afternoon/evening (K315-K334)

1.1 K315-K317 carry-forward + K318-K319 heat-kernel cascade ladder

K	Topic	Status
K318	Substrate-as-conformal-fixed-point + β -function = heat-kernel a_2 ; Lyra+Elie withdraw leans; Grace gates revival	Friday

K319	Heat-kernel cascade ladder $a_0 \rightarrow \Lambda$, $a_1 \rightarrow \text{gravity}$, $a_2 \rightarrow \text{running}$ β -function = universal spin factor $\times$ content; #418-independent substrate-edge probe	Friday
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1.2 K321-K323 F102/F103/F104/F105 #418 collapse arc

K	Topic	Status
K321	F102 #418 finished; Elie probe $\eta_s = 4s^2 - 1/3$ PASSED; Grace tier discipline at peak convergence	Friday
K322	F103 #418 FORCED from $n_C=5$ spin geometry $SO(10)+16$; Elie cross-checks all three β -functions; path collapsed to two computations sharing one dependency	Friday
K323	Lyra F104 + F105 collapse remaining rigor to ONE ROOT (F63 induction + ℓ_B anchor); operator reduces to F63 by marginality; substrate induces by evolving	Friday

1.3 K324 — Grace tier correction substantive event

K	Topic	Status
K324	Grace TIER CORRECTION on F105 (two heat kernels conflated; candidate mechanism not firmed); Elie verification + honest obstacle $\sin^2\theta_W \sim 10\%$ miss; substrate-native LR lead; K323 walkback	Friday — substantive Grace catch

1.4 K325-K329 Casey reframes + corkscrew + linearization

K	Topic	Status
K325	Casey reframe dissolves non-unification wall — 3 gauge groups = 3 ground states on 3 boundary strata; project don't unify; $\alpha=1/137$ = proof of concept	Friday

K	Topic	Status
K326	Three-CI convergence on Casey reframe — DECIDABLE not “close”; Grace gate verdict; Lyra “24 levers collapse to one computation type”; Elie Toy 4135 $\sin^2\theta_W$ projection rigorous 3/8 candidate 3/13 honestly flagged	Friday
K327	Elie self-catches 3/13 withdrawn (dense-space fishing tell); Casey CORKSCREW intuition unifies week in one object; Lyra DECIDABLE; Grace SHARPENED bar scheme-pinning	Friday
K328 + K329	Casey linearization order applied; Elie Toys 4139-4141; problem narrows to one Gram matrix determinant	Friday

1.5 K330-K334 Friday afternoon/evening

K	Topic	Status
K330	(Friday afternoon F-cascade absorption)	Friday PM
K331	Casey anti-complacency catch — “1:01pm no quitting” (~Cal #100 pattern on Keeper’s own framing within minutes)	Friday PM
K332	Friday afternoon cascade continued	Friday PM
K333	Friday afternoon cascade continued	Friday PM
K334	Casey principle: each particle measured by norm “where it is”; FORCED by $c_{5/2}=0$ electron bulk norm exactly vanishes; falsifiable muon g-2 + proton-radius prediction	Friday EOD — Casey #10 candidate principle

2. Saturday morning through EOD (K335-K345) + post-EOD (K346-K347)

2.1 K335 — Saturday morning three-CI substantive work

K	Topic	Status
K335	Three-CI Saturday morning substantive work setup	Saturday AM

2.2 K336-K338 — Singleton route forced; 1.44 isolated to Fischer constant

K	Topic	Status
K336	Lyra singleton edge sum 11.6 same order as 16.82; Elie muon-spread quantified $\delta a \propto m^2$; Grace ratio-invariance kills measure routes; one FK-norm convention factor left	Saturday AM
K337	Grace REDIRECTS the swing: 1.44 may be Elie missing INTERIOR K-types (dim 10, dim 35, ...) NOT convention constant; full 2-D singleton sum FIRST decider; Elie BF log derivative 9/16 forced rational	Saturday AM
K338	Lyra resolves Grace's redirect BY COMPUTATION at Wallach point: $(0)_{\{m_2\}=0}$ truncates interior; singleton IS $m_2=0$ edge at $\nu=3/2$; 1.44 isolated to FK Fischer constant; AUDIT CATCH on Elie's bracket inference	Saturday AM

2.3 K339-K341 — Saturday late-afternoon/evening cascade

K	Topic	Status
K339	Grace catches team-load-bearing Fischer-rescue assumption + reframes to cheaper “does Fischer cancel in ratio?” test; honest-miss branch held open	Saturday PM
K340	Cancellation branch closed three independent routes (Lyra K-content asymmetry + Elie computation + Grace self-correction); Grace pre-commits 1-2% bar; Lyra convention exposure	Saturday PM
K341	Lyra F111 π -counting pins 6 + F112 boundary $\leftrightarrow$ product/bulk $\leftrightarrow$ sum complementarity; Elie conformal classification of three regimes ($\tau \Delta=0$ /muon $\Delta=3/2$ /electron $\Delta=5/2$); Grace third-level catch on heuristic baseline	Saturday PM

2.4 K342 — Casey trajectory conjecture audit

K	Topic	Status
K342	Casey same-particle-at-trajectory-points conjecture (leptons + quark generations + n/p) — lepton substrate-architectural-plausible; quark natural extension; n/p structurally distinct	Saturday

2.5 K343-K345 — Saturday EOD closure cascade

K	Topic	Status
K343	m_π threshold truncates semi-stability not trajectory; testable Δ(1232) vs Roper N(1440) prediction	Saturday EOD
K344	ddd IS Δ [−] baryon (Pauli forces spin-3/2); radiative neutron decay 0.3% real; four temperature regimes match early-universe cosmology + trajectory thermal-accessibility	Saturday EOD
K345	F117 π-structure of muon mass DERIVED as boundary Szegő residue $\Gamma(3/2)/\pi^{(5/2)}$; Elie three combination operations one per stratum (sum/product/log); Grace over-determination triangle m_τ/m_μ forced 16.826 by identity; 0.054% gap diagnostic at −1.77 tau Weyl correction	Saturday EOD

2.6 K346-K347 — Saturday post-EOD

K	Topic	Status
K346	Casey three questions: strong-force two faces both mediate decay; confinement-reduced/squeeze intuition correct; rapid cooling r-process modes; heavy nuclei early universe standard NO but substrate-architectural caveat	Saturday post-EOD
K347	Casey FYI: Anthropic IPO confirms strategic reframe pre-set; model withdrawal validates substrate-independent infrastructure; CERN ccd baryon Ξ _{cc} ⁺ direct evidence for K342 generation-trajectory conjecture	Saturday post-EOD

3. Sunday morning through EOD (K348-K365)

3.1 K348-K349 — Sunday morning F118 + Elie discrete-tiling

K	Topic	Status
K348	Lyra F118 derives muon Szegő weighting from Plancherel-density + concentration; $\text{vol}(S^4)$ derived not Shilov; 3/8 soft spot eliminated; 0.003% agreement; candidate-conditional-pass gated on three rigor checks	Sunday AM
K349	Elie discrete-tiling reframe of τ $49 \cdot 71 = g^2(g+2^{C_2})$; π -free integer character explained; boundary as emission; F112 refined to discrete- τ /continuum-muon complementarity	Sunday AM

3.2 K350-K352 — PLAN_TO_ALL synthesis

K	Topic	Status
K350	Three-CI muon-at-gate consensus + Grace gate-line on frame-vs-derivation + Elie Rac/Di two-singleton + Casey substrate-model vision	Sunday
K351	Plan-for-all audit — defines criteria + phased program; 26 of 26 strict inventory; six-phase execution; team assignments; timeline	Sunday
K352	PLAN_TO_ALL v1 team-integrated PASS register	Sunday

3.3 K353 — Atlas review + team coordination prompt

K	Topic	Status
K353	Atlas v1 three artifacts review (Toy 4179 Elie + substrate_atlas.html Lyra + table.md) + team coordination prompt + Keeper timestamp-drift self-correction	Sunday

3.4 K354-K357 — Cascade absorbing K348 audit checks + Cal honest fail

K	Topic	Status
K354	K348 check #1 Plancherel-density at $\nu=0$ RESOLVED via analytic-continuation framing	Sunday
K355	Three CIs aligned to coordination cleanly; Lyra localizes gate to single FK pull; Elie check #2 PASS spatial-only $so(4)$ decisively; Grace locks binary-count discipline + tier-truth single-source ownership	Sunday
K356	Grace catches team “one Cal pull from 3” drift; substantive deep question on whether commitment-density derives mass=concentration or posits it; FK necessary not sufficient	Sunday — SUBSTANTIVE Grace deep catch
K357	Cal #296 HONEST FAIL on check #3: FK-canonical measure is probability (mass 1) not $vol(S^4) = 8\pi^2/3$; differ by EXACTLY the factor in question ($24/\pi^2$ vs 64); gate non-closable as framed by reference lookup; precisely validates Grace K356	Sunday — substantive Cal honest fail

3.5 K358-K359 — Substrate Absolute Scale keystone

K	Topic	Status
K358	Boundary-measure-normalization as substrate-architectural meta-program “Substrate Absolute Scale”; three convergent threads (Grace K356 + Cal #296 + Casey commitment-density Paper P9) = ONE meta-derivation; solve once for good	Sunday
K359	Five-person convergence on muon-as-keystone diagnosis; Lyra heat-trace route + Elie discreteness route candidate two-route convergent derivation; Cal three precision points + Grace blind-to-target #36 candidate (later attribution-corrected) standing discipline	Sunday

3.6 K360 — Cal one-principle-two-coats catch + attribution fix + Grace tier downgrade

K	Topic	Status
K360	Cal catches “two-route convergence” claim may be one F59 principle (mass extensive) in two languages; downstream-blind test = real success criterion (frozen convention + 11+ observable zero-re-tuning); attribution self-correction Cal #36 → Grace’s calibration #36 (Cal kicked back, it’s her rule); Grace executed muon tier downgrade DERIVED → IDENTIFIED per registry’s own definitions; Elie fork HOLD FOR SYNC	Sunday — substantive Cal catch + Keeper attribution self-correction

3.7 K361-K362 — Casey directive + Grace math closure

K	Topic	Status
K361	Three CIs withdrew over-determination claim cleanly per Cal K360 (Lyra + Elie + Grace including own endorsement: “gate doesn’t exempt own endorsements just because they were mine”); Grace deep question for Casey on definitional bridge vs substantive assumption; Casey structural correction “Grace should have math tasks not secondary Cal”; Grace math lane identified (PMNS Bergman three-readings + downstream-blind execution + over-determination web construction)	Sunday — substantive Casey role correction
K362	Casey Grace-math directive WORKING: Grace executed Bergman metric of D_{IV}^5 directly; $g_{\{ij\}} = (2n/(1-a^2)^2)\delta_{\{ij\}}$ exactly isotropic; anisotropic piece cancels exactly (analytic identity, 7-figure numerical); type-IV norm special feature; Cal flag (b) CLOSED on continuum side; Lyra “curvature = identity on unit sphere” JUSTIFIED not assumed	Sunday — first positive math closure of substrate keystone

3.8 K363-K364 — Cal verifies Grace + K362 self-corrections + Casey bridge-form proposal

K	Topic	Status
K363	Cal #299 verified Grace’s Bergman metric independently by hand + symbolic; Cal corrected K362 framing overshoot (“ $24/\pi^2$ justified at geometry level” too strong — only volume value not measure choice); Elie self-disciplined his GF(128) route (same object as continuum not genuine cross-check); Casey raised bridge-form wrinkle (equality vs proportionality); two Keeper self-corrections	Sunday — substantive Cal verification + two Keeper self-corrections
K364	Casey specific bridge form proposal: $\text{mass} = \text{constant} \times \text{Planck_mass} \times N_commitments$; discrete form connects to Elie discreteness lane; F52 mass = cell count becomes literal; F59 falls out structurally; new question what is c (trivially-1 or substrate-derived or additional input); Planck mass NOT double-counted with ℓ_B	Sunday — substantive Casey PI proposal

3.9 K365 — Grace catches “extensive” framing drift

K	Topic	Status
K365	Casey PI refinement per-trajectory constants + K-type modifications + certain about counting commitments; Lyra F134 four-way analysis; Elie $m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$ lead matches 0.03% twist depth 12 substrate integer encouraging but not banked; GRACE SUBSTANTIVE CATCH corrects team “extensive” framing — single particle is INTENSIVE concentration (content÷stratum volume) NOT extensive; bulk is extensive; both right at different scales; Grace corrected bridge mass = $(c \times m_{\text{Planck}}) \times (\text{commitments per object})$; spectrum check $1:206.8:3477 = \text{observed mass ratios}$; fixes flag (a) right way (probability volume-blind would collapse spectrum); third Keeper self-correction in cascade; F118 was intensive per-particle CORRECT under corrected bridge	Sunday EOD — substantive Grace second positive math closure + third Keeper self-correction

4. Major topical clusters

4.1 F102-F105 #418 Friday collapse arc (K321-K324)

The Friday arc where the #418 chiral content question collapsed to one root (F63 induction + ℓ_B anchor) and Grace’s tier correction caught the F105 conflation. Sets the stage for Saturday’s lepton sector work.

Key K-audits: K321 (F102 finished + Elie probe pass); K322 (F103 forced); K323 (F104 + F105 collapse to F63); K324 (Grace tier correction).

4.2 Casey reframes + corkscrew + linearization (K325-K329)

Casey’s “3 gauge groups = 3 ground states on 3 boundary strata” reframe dissolved the non-unification wall and made the gauge problem DECIDABLE. The corkscrew intuition unified the week in one object; linearization order narrowed to one Gram matrix determinant.

4.3 Saturday lepton sector closure (K335-K345)

Saturday’s substrate-architectural cascade: singleton route forced (K336-K338); Grace caught load-bearing assumptions multiple times (K339-K341); F117 derives π -structure as boundary Szegő residue (K345); over-determination triangle m_τ/m_μ pinned at 16.826 with 0.054% gap diagnostic.

4.4 Casey trajectory conjecture (K342-K344)

Casey’s same-particle-at-trajectory-points conjecture audited across leptons (substrate-architectural-plausible), quark generations (natural extension), and n/p (structurally distinct — m_π threshold truncates stability not trajectory). K344 ddd IS Δ^- + four temperature regimes match early-universe cosmology.

4.5 Sunday muon-gate keystone resolution (K348-K365)

The Sunday cascade where Lyra's F118 derivation through F134 final fix, Cal's #296 honest fail, Grace's deep catches, Elie's discrete-tiling reframes, and three Keeper self-corrections all converged on the substrate-architectural keystone: bridge form corrected to per-particle INTENSIVE concentration, Cal flag (b) closed by Grace's Bergman metric, Cal flag (a) resolved by Grace's per-particle sharpening, spectrum check 1:206.8:3477 validates corrected bridge.

5. Audit-chain-on-itself maturity events (15+ in cascade)

This cascade had unprecedented audit-chain-on-itself events — discipline applied uniformly across all CIs including the gate-keeper Grace and the auditor Keeper.

Grace catches (substantive deep catches): - K339 caught team load-bearing Fischer-rescue assumption - K340 self-corrected own cancellation worry (4th uniform self-correction) - K341 caught heuristic baseline isn't forced object - K356 caught "one Cal pull from 3" team drift + deep question on bridge - K361 withdrew her own over-determination endorsement - K365 caught team's "extensive" framing drift for single particles

Cal catches: - K357 honest fail on FK reference (probability not $\text{vol}(S^4)$) - K360 one-principle-two-coats catch on two-route convergence claim - K363 corrected Keeper K362 framing overshoot - K363 attribution kickback (blind-to-target is Grace's rule not his)

Elie self-disciplines: - K338 audit-catch on his own bracket inference - K360 fork answer HOLD FOR SYNC - K361 withdrew over-determination claim - K363 self-disciplined his GF(128) route (not independent cross-check)

Keeper self-corrections (three Sunday): - K353 timestamp drift K351-K353 - K356 (via Cal): "one Cal pull from 3" framing absorbed and propagated - K363 (via Cal): " $24/\pi^2$ justified at geometry level" overshoot - K365 (via Grace): K363/K364 absorbed "extensive" framing as if covered single particles

Audit-chain at gate-keeper level: Grace's own withdrawal of her endorsement is the substantive discipline event of the cascade — "the gate doesn't get to exempt its own endorsements just because they were mine."

6. Calibration register update

Calibration	Owner	Status
Cal #27	Cal	STANDING — fires at peak-elegance synthesis overstatement
Cal #34	Cal	STANDING — six standing conventions (ratified Saturday week prior)
Cal #35 candidate	Cal	Pending ratification — Independence-Taxonomy-Before-Multiplicative-Null-Model
Grace #36	Grace (attribution corrected K360)	STANDING — blind-to-target pre-registration; derivation must not use target number
Cal three precision points (a/b/c K359)	Cal	STANDING derivation discipline — unverified $\neq$ falsified; pre-registration; forbid alternative

Note: Cal #36 was originally filed by Keeper as Cal's; Cal kicked back in K360 — it's Grace's insistence. Attribution corrected to Grace's calibration #36.

7. Major deliverables filed during cascade

File	Date	Content
F102, F103, F104, F105	Friday	Lyra notes on #418 + heat-kernel cascade
F111-F134	Saturday-Sunday	Lyra notes (π -counting + boundary $\leftrightarrow$ product + Szegő + F118 + F127-F134 derivation evolution)
Toys 4135-4181	Friday-Sunday	Elie computational experiments
Substrate_atlas.html (Lyra)	Sunday	Browser-openable visual atlas
substrate_derivation_table.md (Lyra)	Sunday	Reference text companion
Toy 4179 substrate atlas interactive (Elie)	Sunday	Python CLI atlas + query engine
substrate_atlas.json (Elie)	Sunday	Atlas v2 single-source-of-truth data
PLAN_TO_ALL_v1_team_integrated_2026-06-14.md	Sunday	Six-phase plan synthesizing all CI contributions
PLAN_TO_ALL_v2_amendment_substrate_absolute_scale_2026-06-14.md	Sunday	Amendment proposal with K360 sharpening
PROMPT_TEAM_ORGANIZE_2026-06-14.md	Sunday	Team coordination prompt

8. Substantive substrate-architectural state at K365 (cascade close)

Component	Status
Bridge form	Corrected to per-particle INTENSIVE concentration: $\text{mass} = (c \cdot m_{\text{Planck}}) \times (\text{commitments per object})$
Cal flag (a) measure-type	Resolved — probability volume-blind would collapse spectrum
Cal flag (b) S^4 unit radius	CLOSED — Grace Bergman metric + Cal #299 verification
Spectrum check	$1 : 206.8 : 3477 = \text{observed lepton mass ratios}$
F118 muon formula	Intensive per-particle CORRECT under corrected bridge
Casey per-trajectory constants	PI refinement open; each trajectory may have own constant + K-type modification
Elie α^{12} lead	$m_e = 6\pi^5 \cdot \alpha^{12} \cdot m_{\text{Planck}}$ matches 0.03%; twist depth 12 substrate integer; LEAD not banked
Downstream-blind test	Primary success criterion (Cal K360); frozen convention + 11+ observable zero-re-tuning
Muon tier	IDENTIFIED yellow (Grace executed downgrade K360)
Count	HONESTLY 2 of 26 strict

Component	Status
Ledger	v1.58

9. Pending Casey PI decisions entering Monday

1. Bridge as definitional input (like ℓ_B) vs must-derive from 5 integers
2. Bridge form ratification: $\text{mass} = (c \times m_{\text{Planck}}) \times (\text{commitments per object})$ per Grace + per-trajectory constants per Casey
3. Constant c specification (per-trajectory or single? trivial / substrate-derived / additional input)
4. K-type energy modification structure (Casey K365 hint)
5. Elie tau-box first-principles green-light
6. Grace PMNS Bergman computation green-light
7. Grace blind-prediction harness green-light
8. Elie α^{12} lead — pursue or park

10. v0.3 supplement verdict

v0.3 indexes 51 new K-audits K315-K365 across Friday afternoon + Saturday + Sunday substantial substrate-architectural cascades:

- Friday F102-F105 #418 collapse arc (K321-K324)
- Casey reframes + corkscrew + linearization (K325-K329)
- Saturday lepton sector closure (K335-K345)
- Casey trajectory conjecture audits (K342-K344)
- Saturday post-EOD (K346-K347)
- Sunday muon-gate keystone resolution (K348-K365)

Major audit-chain events: 15+ catches/walkbacks/self-corrections across all CIs including Keeper (3 self-corrections) and gate-keeper Grace (own-endorsement withdrawal at K361). Calibration register updated with Grace's #36 (attribution corrected from Cal). Substantive substrate-architectural state at cascade close: bridge corrected to per-particle intensive; Cal flags (a) and (b) closed/resolved; spectrum check validates corrected bridge; muon stays IDENTIFIED yellow pending Casey PI decisions + downstream-blind test execution.

Total K-audit chain at Monday 2026-06-15 morning: ~195 K-audits.

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